

Chapter 2 — Evaluation of the Urologic Patient: History and Physical Examination

- **History and examination come first;** laboratory and imaging tests are chosen on their findings to narrow the differential.
- **A reproducible, organised system** identifies cause, contributing factors and information relevant to treatment, and **builds the rapport** needed for later counselling.
- **Diagnostic confirmation bias:** relying on prior records, "chart lore", aberrant laboratory values or incidentalomas without the patient's input misdirects evaluation — e.g. a haematuria referral whose penile tumour is found only on examination.

Concept — *The history and examination are the primary diagnostic test; prior data inform them but never replace them.*

HISTORY

- **Foundation of evaluation:** a good history often gives the diagnosis or directs to it.

Setting

- **Warm, non-threatening environment** with easy intake — frustration with parking or front-office staff agitates the patient before the encounter.
- **Gender-sensitivity training for all staff:** preferred name and pronouns.
- **Review vital signs and records before entering;** knock and introduce yourself.
- **Face-to-face layout,** computer to the side, so the patient does not feel secondary to it.
- **Non-verbal communication conveys most emotion and affect;** eye contact and posture alter disclosure, and **sitting rather than standing** improves satisfaction and outcomes and reduces litigation.

Comprehension

- **Plain language;** health literacy is often misjudged across a language barrier, and **patients may decline interpreters out of embarrassment.**
- **Teach-back method** — the patient restates the information in their own words — confirms and reinforces understanding.

Companions

- **A companion** supplies history the patient lacks and, since **patients stop listening effectively after difficult news,** can take notes and relay information once the shock passes; also aids recall of long risk-benefit discussions.

Components

- **Complete history:** chief complaint, history of present illness, past medical and surgical history, allergies, social and family history, review of systems.
- **Direct, focused questioning:** let the patient voice concerns, then steer toward what diagnosis and safe treatment require.

Chief Complaint

- **Why the patient came, and the urologist's primary focus.**
- **Address it even when a more serious problem is found:** a patient with frequency and an incidental renal mass regards care that treats only the mass as ineffective.
- **Start the differential from the complaint;** narrow it with the history of present illness.

History of Present Illness

- **Timing, severity, nature, and exacerbating and relieving factors** of the chief complaint; an efficient history of present illness (HPI) narrows the differential diagnosis.

Pain

- **Location, nature, onset, duration, prior episodes**; the patient **points with one finger** to the maximal site.
- **Stone-naïve patients misattribute renal colic** to gut or musculoskeletal causes; previous stone formers recognise it and present promptly.
- **Pain versus tenderness**: pain is reported; tenderness is elicited by palpation. **Pain without tenderness suggests referred pain** — testicular pain with a non-tender testis may come from an ipsilateral obstructing ureteric stone rather than orchitis.
- **Severity**: mild–moderate–severe or 0–10; the **Wong–Baker FACES scale (1988)** uses faces of increasing distress; record before and after intervention.

OBSTRUCTIVE VERSUS PARENCHYMAL PAIN

Feature	Obstructive (colicky)	Parenchymal (constant)
Cause	Distension of a hollow organ (e.g. pelviureteric junction stone)	Inflammation, infection or subcapsular bleeding stretching the renal capsule (e.g. pyelonephritis)
Pattern	Intermittent	Constant
Behaviour	Restless , moving to find a position of comfort	Lies still , since movement worsens pain

- **Shared innervation explains associated features**: the coeliac plexus supplies both foregut and kidneys, so **renal irritation causes paroxysmal nausea and vomiting**; the ureter shares its nerve supply with the ipsilateral testis or labium, so ureteric pain is referred there.
- **Pain from tumour infiltration of periparenchymal nerves** is rare and indicates advanced disease.

Acute pain

- **Clear cause**: obstruction or inflammation → prostaglandins and chemokines → nerve stimulation → central nervous system, with amplification or dampening en route.
- **Analgesia acts at either end**: block the mediator (non-steroidal anti-inflammatory drugs [NSAIDs] inhibit prostaglandin synthesis) or act on central opioid and other receptors.

Renal pain

- **Ipsilateral costovertebral angle**, just lateral to the spine and below the 12th rib.
- **Collecting-system obstruction → colicky pain; parenchymal inflammation or infection → flank pain with costovertebral angle tenderness (CVAT)**.
- **Radiates** anteriorly across the flank toward the midline, or down to the ipsilateral scrotum or labium.
- **Gut and musculoskeletal causes mimic it**:

RENAL VERSUS INTRAPERITONEAL PAIN

Feature	Renal	Intraperitoneal
Relation to food or bowel function	Absent	Typical
Site of maximal tenderness	Costovertebral angle	Anterior (e.g. Murphy sign in cholecystitis)
Peritoneal signs	Absent	Exquisite tenderness with any abdominal motion
Ipsilateral shoulder-tip pain	Typically absent	May occur — diaphragmatic irritation via the phrenic nerve

Ureteric pain

- **Acute obstruction: immediate onset, ipsilateral lower quadrant.**
- **Mechanism:** acute distension and hyperperistalsis → prostaglandin accumulation → ureteric spasm and lactic acid production → irritation of A-delta and C fibres → **T11–L1 dorsal root ganglia** → pain perceived in those segments.
- **Gradual or partial obstruction may be painless.**
- **Referred to the ipsilateral scrotum or penis; ureterovesical junction obstruction causes irritative symptoms** — urgency and frequency with small voids.

Vesical pain

- **Suprapubic pain** from inflammation (cystitis, interstitial cystitis) or distension (obstruction, detrusor failure).
- **Inflammatory pain worst when full, eased by voiding.**
- **Strangury:** stabbing pain at the end of micturition from final contraction of an inflamed detrusor.
- **Acute retention in a sensate bladder** is a history diagnosis — intense desire, inability to void. **An atonic bladder retains large volumes painlessly**, as chronic ureteric obstruction is painless.

Prostatic pain

- **Deep pelvic, poorly localised, mistaken for rectal pain;** with irritative symptoms; acute or chronic, inflammatory and/or infective.

Penile pain

- **Flaccid penis:** balanitis, paraphimosis, ulcerative lesions (carcinoma, herpes), or pain referred from cystitis or prostatitis. **Erect penis:** Peyronie disease or priapism.

Scrotal pain

- **Skin causes** (e.g. an inflamed pustule from an ingrown hair) versus **testis and cord.**
- **Epididymitis and orchitis: pain may be relieved by testicular elevation.**
- **Torsion of the testis or its appendages → acute vascular congestion; testicular torsion is a surgical emergency.**
- **Varicocele: dull ache worse at day's end** from venous congestion; eased by elevation and lying supine.
- **Renal and ureteric pain refers to the ipsilateral scrotum** (shared embryology and nerve pathways).

Chronic pain

- **Central sensitisation** (signal "imprinting"): pain without noxious stimulus — chronic pelvic pain, fibromyalgia. **No urological cause after thorough evaluation → pain specialist.**

Concept — *Pain character localises pathology: colic means a distended hollow viscus, constant pain a stretched capsule, and pain without tenderness referral along a shared nerve supply (T11–L1, coeliac plexus).*

Opioid Abuse Considerations

- **United States opioid epidemic:** a national public health emergency was declared on 26 October 2017.
- **Diversion:** exchanging prescribed controlled substances for money or illicit drugs.
- **Physicians and manufacturers implicated:** synthetic opioids falsely promoted as "safer"; complacent overprescribing.
- **Treat the painful condition promptly; prefer non-opioid analgesia** — NSAIDs, physical therapy, neuromodulation, acupuncture.
- **Pain without a urological cause → no opioids;** refer to primary care or pain management.
- **Opioid adverse effects:** euphoria, constipation or ileus, dizziness, nausea, vomiting, tolerance, physical dependence, respiratory depression.

- **Drug-seeking signs:** unusual home locations; inconsistent behaviour (walks easily in the corridor, struggles in the room); poor follow-up; disinterest in non-opioid analgesia; brand-name requests; discrepant or rehearsed answers.

Haematuria

- **Up to a quarter of patients presenting with haematuria have a urological cancer**, so it must always be evaluated; it is gross or microscopic.

Gross haematuria

- **Malignancy risk factors to elicit:** age, irritative voiding symptoms, smoking, industrial chemical exposure; also alkylating chemotherapy, analgesic abuse, chronic foreign bodies in the urinary tract and family history of malignancy.
- **First exclude** urinary tract infection (UTI), nephrological disease, trauma and recent instrumentation; then **full evaluation per American Urological Association (AUA) guidelines**. Age is a risk factor, but genitourinary (GU) malignancy occurs in children.
- **Commonest cause of gross haematuria over 50 years: bladder cancer.**
- **Pseudohaematuria** (dehydration, drugs, foods): **differentiate by microscopy of a clean midstream specimen for red blood cells (RBCs); dipstick alone is insufficient** — other solutes read positive, and evaluating dipstick-only "haematuria" is costly and low-yield.

TIMING OF HAEMATURIA WITHIN THE STREAM

Portion of stream	Suggests
Initial	Mild bleeding from a urethral or prostatic source
Total	Source at or above the bladder
Terminal	Bladder neck irritation , expressed as the bladder neck contracts at the end of voiding

- **Usually painless;** pain arises when clots obstruct the upper tract (colic) or cause retention.
- **Clot shape: vermiform → upper tract; cuboid → bladder.**
- **Response to a catheter:** clears with bladder irrigation but recurs → **upper-tract source**; clears once a catheter is placed → **prostatic source**.

Microscopic haematuria

- **AUA definition: ≥ 3 RBCs per high-power field on microscopy of a properly collected midstream specimen;** a positive dipstick is confirmed by microscopy.
- **Patients on anticoagulant or antiplatelet therapy are evaluated in the same way** as those who are not.
- **Treat benign causes (gynaecological, UTI), then repeat urinalysis** for persistence; **likely intrinsic renal disease → nephrology.**

RISK-STRATIFIED EVALUATION OF MICROSCOPIC HAEMATURIA (AUA)

Risk group	Evaluation
Low	Shared decision: surveillance versus work-up. Work-up = cystoscopy + renal ultrasound. Surveillance = repeat urine microscopy; positive repeat → reclassify as intermediate risk
Intermediate	Cystoscopy + renal ultrasound
High	Cystoscopy + multiphasic computed tomography (CT) urogram (or other axial upper-tract imaging)

- **Cystoscopy can often be performed in the office.**

Concept — Haematuria evaluation **escalates with malignancy risk:** confirm true haematuria by microscopy, remove benign confounders, then match the intensity of upper-tract imaging to risk, while cystoscopy is constant for anyone investigated.

Lower Urinary Tract Symptoms

- **Lower urinary tract symptoms (LUTS):** symptoms of the bladder and its outlet, from any combination of obstructive and irritative causes.
- **Consider a hypocontractile or acontractile bladder** (e.g. diabetes). **Bladder outlet obstruction (BOO) causes a spectrum** from raised residual to complete retention.
- **No single post-void residual (PVR) volume defines retention;** judge the PVR trend together with bother, bladder stones, haematuria, infection and renal injury.
- **Slow progression** (e.g. **benign prostatic hyperplasia [BPH]**) and **neuropathy** make patients under-appreciate obstruction despite objectively weak flow and high residual. **Chronic BOO itself causes detrusor irritability.**
- **Bladder diary:** timed intake and output volumes, with urgency and leakage episodes; **usually 48 hours;** gives **functional capacity (normal adult 300–400 mL)** and quantifies nocturia.

Irritative symptoms

- **Frequency, urgency and dysuria;** causes besides chronic BOO: overactive bladder, cystitis, prostatitis, bladder stones, bladder cancer.
- **Frequency:** more than five or six voids a day; the diary separates **incomplete emptying, overactive bladder and polyuria** (polydipsia, diabetes mellitus [DM], diabetes insipidus).
- **Urgency:** difficulty postponing micturition — normal only after prolonged holding. **Dysuria:** painful voiding from bladder inflammation, felt along the urethra or at the meatus.
- **Irritative symptoms may be the only sign of occult bladder cancer, particularly carcinoma in situ (CIS)** — especially over 50 years and in smokers → **low threshold for cystoscopy.**
- **Nocturia:** number of times the patient wakes to void. **Daytime frequency without nocturia** → **psychogenic** (e.g. anxiety). **Nocturia without daytime frequency** → **nocturnal polyuria:** evening fluid intake; mobilisation of leg oedema when recumbent; age-related loss of concentrating ability with higher nocturnal renal blood flow.

Obstructive symptoms

- **Causes — anything narrowing the urethra:** BPH, prostate cancer, urethral stricture, dysfunctional voiding, detrusor–external sphincter dyssynergia, severe phimosis or meatal stenosis.
- **Complete obstruction** → **acute retention;** **partial obstruction** → **obstructive LUTS:** incomplete emptying, frequency (more often than 2-hourly), intermittency, weak stream, straining.
- **Weak detrusor + minimal obstruction can still cause obstructive LUTS or retention;** a weak stream must be assigned to outlet or detrusor.
- **Hesitancy:** flow normally starts about a second after sphincter relaxation; delayed in BOO. **Intermittency:** stop–start stream, distinct from double voiding.
- **Post-void dribbling:** urethral urine not "milked back" into the bladder; **often an early BPH symptom.**
- **Straining:** Valsalva or manual pressure (Credé voiding) — temporising until definitive treatment.

AUA Symptom Score and quality-of-life score

- **AUA Symptom Score (1992):** seven symptoms over the past month, each 0–5 (Table 2.1). **International Prostate Symptom Score (IPSS) = AUA score + a quality-of-life (QoL) question.**
- **Total 0–35:** mild 0–7, moderate 8–19, severe 20–35.
- **Limitations:** non-specific; sixth-grade reading level; unsuitable for some neurological patients. **Value:** standardised comparison before and after intervention.

TABLE 2.1 INTERNATIONAL PROSTATE SYMPTOM SCORE

Item	Question (over the past month, how often...)	Score
1 Incomplete emptying	Sensation of not emptying the bladder completely after voiding	0–5 *
2 Frequency	Voiding again less than 2 hours after finishing	0–5 *
3 Intermittency	Stopping and starting several times during voiding	0–5 *
4 Urgency	Finding it difficult to postpone voiding	0–5 *
5 Weak stream	Weak urinary stream	0–5 *
6 Straining	Pushing or straining to begin voiding	0–5 *
7 Nocturia	Times typically up to void between going to bed and rising	By count: 0 = none ... 5 = ≥5 times
Quality of life	Feeling if the urinary condition stayed as it is now for life	0 delighted → 6 terrible

* 0 not at all; 1 less than 1 time in 5; 2 less than half the time; 3 about half the time; 4 more than half the time; 5 almost always. From Cockett, Aso and Denis — Proceedings of the Second International Consultation on BPH, Paris, 1993 (published 1994).

- **Six items score how often, nocturia how many times; QoL (0–6) measures bother and sits outside the 0–35 total.**

Urinary incontinence

- **Involuntary loss of urine** when bladder pressure exceeds urethral resistance, or the outlet is bypassed.

CATEGORIES OF URINARY INCONTINENCE

Type	Mechanism	Hallmarks
Continuous	Outlet bypassed — urinary fistula (vesicovaginal, ureterovaginal) or ectopic ureter	Constant perineal wetness, independent of urge or effort; ask about gynaecological surgery, radiation, traumatic childbirth
Overflow (paradoxical)	Volume exceeds capacity → intravesical pressure exceeds outlet resistance	Often at night (less guarding); men with long-standing BOO, often unaware of it; cured by relieving the obstruction
Stress (SUI)	Sphincter weakness ; leak with raised intra-abdominal pressure	Cough, sneeze, laugh, lift; multiparous and postmenopausal women; men after radical prostatectomy or aggressive transurethral resection of the prostate (TURP)
Urge	Leak coincident with urgency	Severe overactive bladder, cystitis, neurogenic or poorly compliant bladder; may signal occult bladder cancer
Mixed	Urge + stress	Treat both ; treating one alone may worsen symptoms
Enuresis	Incontinence during sleep	Normal to 3 years; persists in about 15% at 5 years and about 1% by 15 years
Functional	Intact outlet; cannot reach toilet in time	Limited mobility or access; in the elderly, cognitive decline and loss of social inhibition — possibly volitional voiding
<i>Pseudoincontinence (mimic)</i>	No true urinary loss	Chronic vaginal discharge; severe labial fusion trapping urine in the vagina — found on pelvic examination

- **Ectopic ureter**: the typically dysplastic upper-pole moiety of a duplex kidney drains into the vagina or perineum, beyond the external sphincter → **failed toilet training in childhood; continuous daytime wetting in adults, less at night**, because urine pools in the dependent, saccular upper pole when recumbent.
- **Guarding**: voluntary pelvic floor contraction against leakage, in any type.

- **Stress versus urge treatment:** SUI → **increase outlet resistance** (Kegel pelvic floor exercises, urethral sling, artificial urinary sphincter, bulking); **urge** → **relax the bladder** (anticholinergics, intravesical botulinum toxin, neuromodulation).
- **Primary enuresis** (always present) **persisting beyond 6 years** → **urological evaluation**; in a girl whose wetting is continuous by day as well as night, consider ectopic ureter. **Secondary enuresis** (onset after a dry period) **is associated with child abuse and bullying**.

Concept — Every incontinence type is **a pressure-resistance imbalance or a bypass**: stress lowers resistance, urge and overflow raise bladder pressure, continuous bypasses the outlet, and functional incontinence has neither — it is a problem of access.

Sexual Dysfunction

- **Common in women with LUTS**; in men "impotence" is often a symptom of another problem.

Erectile dysfunction

- **Definition (AUA; National Institutes of Health [NIH] Consensus Conference on Impotence, 1993):** inability to attain and/or maintain penile erection sufficient for satisfactory intercourse.
- **May affect more than half of men over 40.** History identifies the **type** — **vasculogenic, neurogenic, psychogenic, endocrinological, drug-induced** — guiding testing and treatment.
- **Poorly controlled DM** → **neuropathy and/or vasculopathy** (endocrinology referral); **diabetic men are more often hypogonadal, reducing phosphodiesterase-5 (PDE5) inhibitor efficacy**.
- **Drugs:** antihypertensives, antidepressants, antipsychotics, testosterone-lowering drugs.
- **Psychogenic clues:** situational erectile dysfunction (ED) — erections with one partner but not another, or with visual or manual stimulation — and preserved nocturnal tumescence.
- **ED can herald occult coronary artery disease (CAD) → cardiology consultation.**
- **Questionnaires:** International Index of Erectile Function (IIEF) and the abbreviated five-item IIEF-5, the Sexual Health Inventory for Men (SHIM) — erectile and orgasmic function, desire, intercourse and overall satisfaction.

Loss of libido

- **Multifactorial** — onset, timing, situational nature. **With ED, suspect hypogonadism** (primary or secondary): **early-morning serum testosterone on two separate occasions, at least 2 weeks apart**.
- **Hypogonadal symptoms:** altered sleep, emotional change, reduced strength, weight gain, infertility, ED.
- **Other causes:** depression, medications, severe illness (e.g. cancer).
- **Questionnaires:** Androgen Deficiency in the Aging Male (ADAM), Aging Male Survey, Massachusetts Male Aging Study (MMAS) questionnaire — these correlate poorly with serum testosterone; **the Hypogonadism-Related Symptom Scale has better sensitivity and specificity against serum testosterone**.

Premature ejaculation

DEFINITIONS OF PREMATURE EJACULATION

Criterion	AUA / Sexual Medicine Society of North America (2020)	International Society for Sexual Medicine (ISSM)
Latency	Within about 2 minutes of penetration, or before	Lifelong: within about 1 minute (or before penetration); acquired: clinically significant reduction, often to about 3 minutes or less
Other criteria	Inability to delay; bother	Inability to delay; negative personal consequences

- **Lifelong or acquired?** Many patients have **unrealistic expectations** — elicit their own estimate of intravaginal ejaculatory latency time (IELT). Population median latency is about 5 minutes; sex therapists regard 1–2 minutes as too short, 3–7 minutes as adequate and 7–13 minutes as desirable.

- **First-line: topical anaesthetics, the stop-start method and sexual counselling;** alternatively a **selective serotonin reuptake inhibitor, on demand or daily.**

Loss of ejaculation

- **Low volume or anejaculation:** androgen deficiency, sympathetic denervation, drugs, retroperitoneal, bladder neck or prostate surgery, midline prostatic or seminal vesicle cyst.

Absence of orgasm

- **Orgasm:** euphoria with genital muscular contractions and endorphin release; usually accompanies ejaculation but **can occur without erection or ejaculation** (e.g. after prostatectomy).
- **Causes:** psychogenic, psychiatric medications, impaired pudendal nerve function (e.g. diabetic neuropathy).
- **Management:** neurological evaluation (vibratory testing of penile sensation), hormonal treatment and/or sexual counselling.

Concept — *Sexual complaints are **symptoms, not diagnoses**: the history sorts erectile, libido, ejaculatory and orgasmic failure into vascular, neural, hormonal, drug and psychogenic causes — and ED additionally flags cardiovascular risk.*

Haematospermia

- **Blood in the ejaculate; almost always non-specific inflammation of the prostate or seminal vesicles, resolving spontaneously;** follows prolonged abstinence or prostate biopsy.
- **Persistent beyond several weeks → genital examination, digital rectal examination (DRE), prostate-specific antigen (PSA), cystoscopy and urine cytology** to exclude GU tuberculosis or cancer.

Pneumaturia

- **Gas in the urine,** interrupting flow like urethral flatus.
- **Commonest cause: fistula between bowel and bladder** → screen for diverticular disease, Crohn disease, enteritis, and recent abdominal surgery or radiation.
- **Rarely, gas-forming organisms** → ask about severe UTIs and immunocompromise; urinalysis and culture.

Urethral Discharge

- **Commonest symptom of sexually transmitted infection (STI);** screen for high-risk sexual behaviour.
Bloody discharge → consider urethral carcinoma.

Fevers and Chills

- **Subjective or measured, and by which route** (oral, rectal, auricular, temporal) — accuracy against core temperature matters in children. **Rigors, even without fever, suggest bacteraemia.**
- **Cystitis rarely causes fever; pyelonephritis usually requires fever for the diagnosis.**
- **Fever with chills may signal sepsis** → evaluation, possibly admission; worse in the elderly and immunocompromised.
- **Constitutional ("B") symptoms** — fever, chills, night sweats, anorexia, weight loss, fatigue, lethargy — a term borrowed from lymphoma staging; suggest **advanced inflammatory, infective or malignant disease** (GU tuberculosis, advanced bladder cancer).

Emotional Distress

- **Establish bother:** a patient with nocturia three times nightly who is not bothered, and has no sequelae of obstruction (renal injury, recurrent UTI, bladder stones), may be **monitored rather than treated** — best captured by the IPSS QoL score.
- **Conversely,** a patient distressed by benign haematospermia may choose full evaluation over reassurance.

- **Rapport lets patients voice distress**, so plans can be tailored.

Concept — *In benign conditions **bother, not the symptom itself, decides treatment** — unless objective sequelae (renal injury, infection, stones) override it.*

Medical History

- **Know each comorbidity's severity and treatment compliance**, not just its name — in DM, control, insulin use and neuropathy; ask whether a recent glycated haemoglobin (HbA1c) is available.
- **Sources**: family, primary care notes, the medication list — though drugs have several indications (hydrochlorothiazide for hypercalciuria or hypertension) and adherence varies.
- **Seek undiagnosed disease with treatment implications**: **occult CAD** (no "heart attack", but exertional dyspnoea or chest pain on prompting); **obstructive sleep apnoea** (large collar size; spouse reports snoring, irregular night breathing, daytime fatigue).

Performance Status

- **Functional ability reflects fitness for surgery or chemotherapy.**
- **Activities of daily living (ADLs)**: dressing, eating, ambulating, toileting, hygiene. **Instrumental ADLs**: housekeeping, managing money, preparing meals, shopping, community participation.
- **ADL deficiency predicted complications after percutaneous nephrolithotomy (PCNL) better than the Charlson Comorbidity Index or the American Society of Anesthesiologists (ASA) score.**
- **Eastern Cooperative Oncology Group (ECOG) and Karnofsky scales** (mainly oncological) grade strenuous and non-strenuous activity, self-care, time out of bed and moribund status; they convey overall health to other providers.

Past Surgical History

- **Prior surgery → adhesions and obliterated planes.**
- **Extensive prior abdominal surgery → Hasson (open) or Visiport (optical) access rather than Veress needle** for laparoscopy.
- **Read the operative note, not just its title**: a recorded "TURP" may have been a channel TURP only, so a further TURP may help; notes also reveal anatomy and difficulties encountered.

Medications

- **Reveals disease severity, drug-induced symptoms and perioperative targets.**

Medications with urological side effects

- **Relate the timing of drug starts and stops to symptom onset.**
- **Common culprits**: antihistamines, anticoagulants, antiplatelets, antidepressants, anxiolytics.
- **Examples**: several diabetic drugs and high-dose insulin imply severe DM and probable neuropathy; **tamsulosin → anejaculation, light-headedness; haematuria on antithrombotics → temporary cessation plus full evaluation.**
- **Know the indication**: low-dose aspirin for primary CAD prophylaxis versus prevention of coronary stent thrombosis determines whether and when it may be stopped.

TABLE 2.2 DRUGS ASSOCIATED WITH UROLOGIC SIDE EFFECTS

Urological side effect	Drug class	Examples
Decreased libido; ED	Antihypertensives	Hydrochlorothiazide, propranolol
	Psychotropics	Benzodiazepines
Ejaculatory dysfunction	α -Adrenergic antagonists	Prazosin, tamsulosin
	Central antihypertensive	α -Methyldopa

Urological side effect	Drug class	Examples
Priapism	Psychotropics	Phenothiazines, antidepressants
	Antipsychotics	Phenothiazines
	Antidepressants	Trazodone
Decreased spermatogenesis	Antihypertensives	Hydralazine, prazosin
	Chemotherapy	Alkylating agents
	Drugs with abuse potential	Marijuana, alcohol, nicotine
Incontinence or impaired voiding	Endocrine-active drugs	Antiandrogens, prostaglandins
	Direct smooth-muscle stimulants	Histamine, vasopressin
	Others	Furosemide, valproic acid
Retention or obstructive voiding	Smooth-muscle relaxants	Diazepam
	Striated-muscle relaxants	Baclofen
	Anticholinergics, musculotropic relaxants	Oxybutynin, flavoxate, diazepam
	Calcium-channel blockers	Nifedipine
	Antiparkinsonian drugs	Carbidopa, levodopa
	α -Adrenergic agonists	Pseudoephedrine, phenylephrine
	Antihistamines	Loratadine, diphenhydramine
Acute kidney injury	Antimicrobials	Aminoglycosides, penicillins, cephalosporins, amphotericin
	Chemotherapy	Cisplatin
	Others	NSAIDs, phenytoin
Gynaecomastia	Cardiovascular	Verapamil, digoxin
	Gastrointestinal	Cimetidine, metoclopramide
	Psychotropics	Phenothiazines; tricyclics (amitriptyline, imipramine)

- **Retention follows from either side of the balance:** α -agonists raise outlet tone, while anticholinergics, calcium-channel blockers and antihistamines lower detrusor contractility.
- **α -Blockade links ejaculatory dysfunction and priapism** (prazosin, tamsulosin, phenothiazines, trazodone).
- **Phenothiazines appear under three effects** (ejaculatory dysfunction, priapism, gynaecomastia), and **diazepam under both incontinence and retention.**

Medications with implications for urological surgery

- **Antithrombotic indications:** atrial fibrillation, heart valves, CAD, venous thromboembolism.

HALF-LIVES OF COMMON ANTITHROMBOTICS (BROWN ET AL., 2015)

Drug	Class and action	Half-life
Warfarin	Vitamin K antagonist — blocks synthesis of vitamin K-dependent clotting factors	36–42 h
Heparin	Potentiates antithrombin (low-molecular-weight heparins similar)	1–2 h
Dabigatran	Direct thrombin inhibitor	8–17 h
Rivaroxaban	Direct factor Xa inhibitor	5–12 h
Apixaban	Direct factor Xa inhibitor	12 h
Clopidogrel	Antiplatelet — irreversible P2Y ₁₂ blockade	6 h

Drug	Class and action	Half-life
Ticagrelor	Antiplatelet — reversible P2Y ₁₂ blockade	6–8 h

- **Anticoagulants reduce clotting-factor activity; antiplatelets prevent platelet aggregation;** thrombin inhibitors also reduce thrombin-driven platelet activation.
- **Half-life is not duration of effect for irreversible antiplatelets:** clopidogrel's effect lasts the life of the exposed platelets despite a 6-hour half-life.
- **Weigh thromboembolic risk against procedural bleeding risk;** decide to hold or continue with the cardiologist and/or vascular surgeon and the patient.

BLEEDING-RISK CATEGORIES OF UROLOGICAL PROCEDURES

Risk	Procedures
Low	Most endoscopic procedures — cystoscopy, ureteroscopy, laser lithotripsy
Intermediate	TURP, transurethral resection of bladder tumour (TURBT), robotic pyeloplasty, robotic ureteric reimplantation
High	PCNL, artificial urinary sphincter, robotic nephrectomy

Concept — *Perioperative antithrombotic decisions **balance two risks owned by different specialists:** the procedure's bleeding category and the patient's thrombotic indication.*

Allergies

- **Ask at the first visit; verify the list at every visit.**

Medication allergies

- **Record drug and reaction;** patients confuse allergy with side effects (vancomycin red man syndrome, antibiotic dyspepsia). **Flag true allergies in the electronic medical record (EMR).**

Intravenous contrast allergies

- **Intravenous contrast is often essential for CT and magnetic resonance imaging (MRI),** except for stones and a few other indications.
- **Contrast-allergic patients:** choose ultrasound or MRI; if CT is essential, **premedicate with diphenhydramine and a corticosteroid.** Shellfish or "iodine" allergy is not a specific cross-reactivity with iodinated contrast.
- **Endourology requires intraluminal contrast** to outline anatomy and instruments.
- **Retrograde and antegrade pyelography with iodinated contrast is safe in patients with known iodine, shellfish or contrast allergy,** including prior anaphylaxis, with or without prophylaxis (retrospective cohort).

Social History

Tobacco

- **Chewed, smoked or vaped;** agent and method both matter. Cessation support, including nicotine replacement, is appropriate for chewers too; nicotine derivatives themselves promote malignant change in urothelial and lung cell lines, of unknown long-term clinical significance.

Alcohol

- **Raises cancer risk and liver disease** (wound healing); affects LUTS and sexual function.
- **Admitted heavy drinkers risk life-threatening withdrawal** without prophylaxis — Clinical Institute Withdrawal Assessment for Alcohol (CIWA) protocol or alcohol replacement; **they understate intake,** so anticipate.

Recreational drug use

- **Marijuana, cocaine, opioids, methamphetamines, anabolic steroids**; the agent and the route predict challenges in care, and **intravenous use carries communicable-disease risk**.
- **Stigma causes under-reporting** → build trust, or review prior records.

Sexual relations

- **Do not presume a monogamous, heterosexual relationship**; ask non-accusatory questions ("men, women or both? one partner or several?") to assess STI risk and social support.

Domestic situation

- **Household and nearby support** plan care; **little support** → **poor course after complex surgery**; reveals unacceptable options (a sole carer may refuse long convalescence). **Screen for domestic violence**.

Occupation

- **Worldview, socioeconomic status and industrial exposure to carcinogens**.

Family History

- **Defined inheritance**: autosomal dominant polycystic kidney disease, tuberous sclerosis, von Hippel-Lindau disease, renal tubular acidosis, cystinuria.
- **Familial clustering without a defined pattern**: prostate cancer, urolithiasis.
- **A relative's age at diagnosis can set screening age**.
- **Prostate cancer risk rises with an affected brother, and further with an affected father and brother** (large Swedish population study) — the number of affected relatives matters, not only presence.

Review of Systems

- **Comprehensive system-based checklist**; identifies important problems unrelated to the chief complaint.

KEY POINTS

- Thorough characterisation of symptoms may reveal causes and treatment plans.
- An accurate medical history is critical: comorbidities, medications and surgical history may have urological implications.
- Performance status, assessed by ADLs, helps counsel patients on surgical risk.

PHYSICAL EXAMINATION

- **A complete examination often selects the most appropriate next step**.
- **Offer gender-specific chaperones on request**, given the sensitive nature of GU examination.

Vital Signs

- **Temperature, pulse, blood pressure, respiratory rate and pain score identify the critically ill**; out-of-range values → **immediate physician triage** (all staff have a role); profound hypotension or tachycardia → emergency department.
- **Distress screening as a "sixth vital sign"**, with prompt social work, mental health or pastoral support, reduces waiting-room anxiety.

General Observations

- **Four modes of examination: inspection, auscultation, percussion, palpation**; inspection begins at first contact.
- **Observe** distress, nutrition, dress, strength, mobility, skin, dentition, and **apparent versus stated age**.

STIGMATA OF SYSTEMIC DISEASE ON INSPECTION

Sign	Disease
Jaundice	Advanced liver disease
Buffalo hump, skin striae	Cushing disease
Exophthalmos	Graves disease
Fibrofolliculomas	Birt-Hogg-Dubé syndrome
Cogwheel rigidity	Parkinson disease
Obvious erection	Priapism

Markers of Frailty and Nutritional Status

- **Frailty (American Geriatrics Society):** excess vulnerability to stressors, with reduced ability to maintain or regain homeostasis after a destabilising event.
- **Office tests:** gait speed, get-up-and-go test, grip strength; calf circumference is inversely related to frailty.
- **Dentition reflects health; moist mucosa, hydration. Obese or deformed patients:** rehearse the operative position in clinic.
- **Poor nutrition:** unintended weight loss (wound-healing risk), chronic wounds, temporal wasting; **low albumin, transferrin, prealbumin predict poor surgical outcomes.**

General Examination (Neurologic, Cardiovascular, Pulmonary, Abdominal, Dermatologic)

- **Sets a baseline. Neurological:** mental status, cranial nerves, motor (bulk, tone, coordination, gait), sensory — especially with pain. **Cardiorespiratory:** murmurs, heart failure, pulses, oedema.
- **Abdominal:** four quadrants; **scars are clues in poor historians; examine the painful area last;** light then deep palpation for masses, hepatomegaly, splenomegaly; **peritonitis (rigidity, guarding, rebound tenderness) → urgent transfer to the emergency department.**
- **Skin:** rashes, lesions, chronic wounds → infection, malnutrition, occasionally malignancy.

Urologic-Specific Considerations**Renal**

- **Retroperitoneal**, bounded by psoas, diaphragm, oblique muscles and peritoneum; **normally not visible and difficult to palpate**, except very large masses in small children or very thin adults.
- **The right kidney lies lower than the left (liver);** its lower pole may be felt on deep inspiration.
- **Inspect flank skin** for superficial causes of "flank pain"; test pain (pin or spoke-wheel), temperature (cool object), light touch (brush); **herpes zoster gives prodromal hyperaesthesia before vesicles.**
- **Bimanual palpation (renal ballottement, Fig. 2.1):** non-examining hand posteriorly at the costovertebral angle → examining hand through the anterior abdominal wall → deep inspiration displaces the kidney downward.
- **Neonates and infants:** thumb anteriorly on the abdomen, four fingers posteriorly at the costovertebral angle; **large flank masses in neonates are usually renal.**
- **Transillumination:** fluid (cyst, hydronephrosis) → **dull red glow; solid (tumour, polycystic kidney) → no glow.**
- **Renal bruit** (renal artery stenosis, large arteriovenous fistula) is theoretically audible but **non-specific → Doppler ultrasound or angiography.**
- **Renal "percussion"** tests for pain with the base of a closed fist at the costovertebral angle; **be gentle — a light tap elicits a positive sign.**

Bladder

- **Palpable above the pubis only when it holds more than about 150 mL; visible as a lower midline mass in thin patients at about 500 mL.**
- **Percuss from the symphysis toward the umbilicus:** note dull changes to resonant above the bladder.
- **Ballottement** aids palpation.
- **Bimanual examination (Figs. 2.2 and 2.3)** assesses bladder mobility and is the **standard of care after resection of large bladder tumours**; even in the CT and MRI era it **improves prediction of pT3 disease and is an independent predictor of pT3** on multivariable analysis.

Prostate / Digital Rectal Examination

- **Indications:** size, prostatitis, cancer screening; **bimanual under anaesthesia after TURBT**; **gentle DRE in neonates with differences of sex development** for Müllerian structures (uterus).
- **Screening of average-risk men:** the chapter cites offering screening at **55–69 years every 2 years** (AUA, 2013); the current AUA/Society of Urologic Oncology guideline (2023) advises a **baseline PSA at 45–50 years and regular screening at 50–69 years every 2–4 years**.
- **Technique:** patient standing, feet shoulder-width apart, bent about 90 degrees at the waist, hands or elbows on the table — or lateral decubitus with hips and knees flexed → reassure throughout → non-dominant hand spreads the gluteal folds; inspect the anus for haemorrhoids and lesions → lubricated gloved finger advanced gently, non-dominant hand on the anterior thigh or lower abdomen for counter-traction → advance until the prostate is palpable → provide cleansing supplies and privacy.
- **Normal prostate:** chestnut-sized, rubbery, like the contracted thenar eminence. **Nodular firmness, like a flexed knuckle → biopsy.**

Perineum

- **Anus to scrotal base;** access to the posterior urethra and **transperineal biopsy**. Perineal pain may be prostatitis; inspect for infection and fistula; **tenderness may mean a tight pelvic floor.**

Genitourinary Examination

Phallus

- **Inspect hair distribution, skin lesions, nodules, masses and foreskin; record Tanner stage in children.**
- **Lesions:** superficial vesicles → herpes simplex; ulcers → other sexually transmitted diseases or penile squamous cell carcinoma; warts (condylomata acuminata) → human papillomavirus (HPV). **Prominent superficial veins are usually normal.**
- **Retract the foreskin: most penile tumours involve the prepuce or glans.**
- **Meatus:** position, stenosis and discharge (characterise and sample if STI is suspected). **Hypospadias** (ventral ectopic meatus): record site and calibre (e.g. megameatus). **Epispadias** (dorsal meatus): rare, usually with major anomalies such as bladder exstrophy.
- **Palpation:** dense subcutaneous plaques on the underlying fascia → **Peyronie disease** (with a history of curvature on erection); a tender urethra → periurethritis; **spongiofibrosis is palpable only when severe** (relevant to urethral reconstruction).

Scrotum and testes

- **Scrotal skin:** infections common (tinea cruris, cellulitis, pustules) — erythema and tenderness, or fluctuance and pus in an abscess; **malignancy rare (squamous cell carcinoma, classically in chimney sweeps).**
- **Neglected or immunocompromised → Fournier gangrene:** necrotic black skin, foul odour, "dishwater" discharge, **pain out of proportion to examination, crepitus.**

- **Lie:** vertical and of similar size, one slightly lower. **Unilateral pain with a foreshortened cord and horizontal lie → suspect torsion.**
- **Cremasteric reflex:** light touch to the inner thigh → cremaster contraction elevating the ipsilateral testis. Its absence is highly sensitive for torsion in children; a preserved reflex argues against torsion but does not exclude it.
- **Palpate the normal testis first:** firm, rubbery, smooth and ovoid; **normal adult length about 4–5 cm and width about 2–3 cm**, varying with ethnicity; measured best with an **orchidometer**. Glide the testis between the fingers of both hands to feel its contour.
- **Small testis** → prior infarct, surgery, hypogonadism or endocrinopathy such as Klinefelter syndrome.
- **Tender testis** → **orchitis**; **tender epididymis** → **epididymitis**.
- **Testicular mass:** hard, obliterates the smooth contour → **cancer until proven otherwise**. **Epididymal mass:** obliterates the posterior ridge → **almost always benign**.
- **Vas deferens:** palpable bilaterally, high in the scrotum, like thick al dente linguine.
- **Transilluminate scrotal masses** for fluid (e.g. hydrocele).
- **Varicocele ("bag of worms"):** examine supine, standing and standing with Valsalva.
- **Hernia (Fig. 2.4):** index finger over the testis invaginates the scrotum up to the external ring → other hand over the internal ring and Hesselbach triangle (inferior epigastric artery, inguinal ligament, lateral border of rectus abdominis) → Valsalva → **a hernia is a distinct bulge descending onto the fingertip**. **Children:** **"silk glove" sign** — the layers of the sac roll over each other like the fingers of a silk glove.

Concept — *Scrotal examination is a series of paired contrasts* — *testis versus epididymis, vertical versus horizontal lie, glow versus no glow* — *each pairing separating the benign from the urgent.*

Female Genitourinary Examination

Pelvic examination

- **Indications:** pelvic organ prolapse, urinary incontinence, dyspareunia, blood per urethra or vagina, anterior vaginal masses.
- **Sequence:** undress from the waist, gown → stirrups flex and abduct the thighs and flex the knees → chaperone offered (for all pelvic examinations, male or female) → inspect genitalia and introitus (atrophy, erosions, ulcers, discharge, warts) → separate labia; inspect urethra (prolapse, caruncle, hyperplasia, cysts) → with a full bladder, Valsalva to elicit SUI → half-speculum shows one vaginal wall; Valsalva shows prolapse of bladder, apex or rectum → palpate urethra (mass or expressed discharge → **urethral diverticulum**) → bimanual: two fingers of the dominant hand in the vagina (one if the introitus is small), non-dominant hand over the lower abdomen.
- **Performed on clinical suspicion, not for screening** (screening → gynaecologist); **children, adolescents and young women rarely need a urological pelvic examination.**

Urethral mobility

- **Lack of urethral support, especially at the mid-urethra, is the key factor in SUI.**
- **Q-tip (cotton-swab) test:** dorsal lithotomy → lubricated swab gently to the bladder neck → Valsalva → **deflection of 30 degrees or more from the horizontal indicates urethral hypermobility.**
- **A positive Q-tip test is not required to diagnose SUI.**

Prolapse

- **Symptoms range from none** to pelvic or back pressure, pain, a bulge, change in SUI, and difficulty voiding or defecating.
- **Sequence:** dorsal lithotomy → cough or Valsalva: note leakage and obvious prolapse → palpate levator ani and obturator internus for pain and voluntary recruitment (Fig. 2.5, after Meister et al., 2019) → speculum

examination of anterior, posterior and apical compartments; note presence or absence of the cervix → withdraw the speculum slowly during Valsalva to show maximal prolapse → document (Pelvic Organ Prolapse Quantification [POP-Q] system) → inspect the vaginal epithelium for atrophy, which worsens prolapse.

Neurologic Examination

- **A sensory dermatome map (Fig. 2.6) localises neurological deficits; genital and perianal sensory loss indicates a lesion of the sacral nerves or roots.**
- **Bulbocavernosus reflex tests the S2–S4 spinal reflex arc:** squeezing the glans or clitoris → immediate anal sphincter contraction, felt on DRE; **or tug an indwelling Foley catheter.**

Concept — S2–S4 serve genital and perianal sensation, the bulbocavernosus reflex and continence together, so sacral sensory loss and an absent reflex localise the same lesion.

Considerations for Special Populations

Paediatrics

- **Infants to young adults (sometimes up to 26 years). Toddlers:** history from parents; **adolescents:** parental input plus private questioning on alcohol, smoking, drugs, sex.
- **Non-threatening:** many avoid white coats; family help with examination; **cold or painful manoeuvres last.**

Elderly

- **Carer input valuable, privacy preserved;** low tables, Hoyer lifts, blankets and pillows.
- **Pseudodementia:** depressed cognition and affect from illness → assess baseline function and consider occult illness.

Obese

OBESITY CLASSES (CENTERS FOR DISEASE CONTROL AND PREVENTION)

Class	Body mass index (BMI), kg/m ²
Obesity	≥30
Class 1	30 to <35
Class 2	35 to <40
Class 3 (severe)	≥40

- **Screen for comorbidities:** hypertension, hyperlipidaemia, type 2 DM, thyroid disease, sleep apnoea — untreated, they raise postoperative complications and worsen outcomes.
- **Elective surgery that can be delayed → dietitian-supported weight loss.**

Transgender and Gender Non-Binary People

- **About 5 per 1000 people in the United States;** stigma deters care → compassionate care, cultural humility, no preconceptions.

TERMINOLOGY

Term	Meaning
Sex	Chromosomal or gonadal sex, assigned at birth
Gender (identity)	A person's internal sense of self
Transgender female	Assigned male at birth; female gender identity
Transgender male	Assigned female at birth; male gender identity
Transmasculine / transfeminine	Direction toward a gender identity, not full conversion
Non-binary	Does not identify with a single gender; use the name, "the patient" or they/them

Term	Meaning
Cross-dressing	Wearing clothing of the other sex for entertainment, self-expression or sexual pleasure
Sexual orientation	Independent of gender identity

- **Office:** staff educated in terminology and free of judgement; **EMR asks a two-step question — gender identity and sex assigned at birth**; gender-neutral bathrooms.
- **Examination:** note secondary sexual characteristics; ask what steps have been taken (clothing, grooming, hormones, gender-affirming surgery); **examine breasts and genitalia only for a purpose, as in cisgender patients**; explain anatomy and each step; build trust first.
- **Adjuncts:** a benzodiazepine before the examination for anxiety; **topical oestrogen for 1–2 weeks before pelvic examination to counter the effect of exogenous androgens.**

PATIENT PRIVACY AND CONFIDENTIALITY

- **Privacy and confidentiality are critical to the physician–patient relationship.**
- **Health Insurance Portability and Accountability Act (HIPAA):** enacted to protect people who lost or changed jobs; now recognised as the law enforcing privacy of health information.

TELEMEDICINE OPPORTUNITIES AND LIMITATIONS

Improved Access to Care

- **Pre-pandemic barriers:** reimbursement, regulation, resistance, technology; **COVID-19 distancing accelerated adoption** — urologists' telehealth use rose sharply 2019–2020 (AUA Census).
- **Benefits:** underserved and low-socioeconomic patients; shorter waits, less travel, shorter visits; proxies join remotely; sustained satisfaction.

Limitations

- **No examination or in-person tests** (vital signs, cystoscopy, PVR); needs good connectivity and a HIPAA-compliant EMR platform simple enough for non-technical patients.
- **No assistant pre-gathers results or questionnaires; convenience increases no-shows and late cancellations.**
- **Billing:** Centers for Medicare & Medicaid Services (CMS) waivers expired with the end of the federal COVID-19 public health emergency; continued insurer reimbursement is required.
- **Licensure is state-based:** physicians usually need a licence in the patient's state of residence; pandemic relaxations lapsed, though some states added telehealth licences or exemptions; the **Interstate Medical Licensure Compact** (37 states) grants several licences through one application.

KEY POINTS

- Telemedicine has been associated with improved appointment wait times and continued patient satisfaction.
- A telehealth platform may be challenging for elderly or non-technologically adept patients.

Concept — *Telemedicine trades examination for access: it suits history-driven care but cannot replace a visit where examination or office testing decides management.*

CONCLUSION

- **History and examination detect problems that laboratory tests and imaging cannot;** a broad differential diagnosis directs the HPI; awareness of the needs of elderly, paediatric and transgender patients extends urological care to a wider population.